

IBM University Engagement Project Submission

“Exploring the Power of Agentic AI with IBM Granite and LangFlow”

Project Tile – AI-Powered Investment Analyst

Domain of Project – Finance / FinTech

Student Name	Recent Passport Photo	Email ID	Mobile number [WhatsApp]
Sujal Bhagat		sujalbhagatcodes@gmail.com	8329014204

Problem Statement -

Problem Statement No. 4 – Sequential Task Agent: Investment Analyst

The Challenge

Investment analysts rely on multiple sources of financial information such as company financial statements, stock market performance, quarterly earnings, analyst reports, economic indicators, and financial news. Since this information is distributed across different platforms, analysts spend significant time collecting, validating, and interpreting the data before making investment decisions.

Manual investment analysis is often slow, repetitive, and susceptible to human error. Furthermore, rapidly changing market conditions require continuous monitoring, making it difficult for analysts and investors to respond quickly to new opportunities or risks.

There is a growing need for an intelligent AI system capable of automating sequential financial analysis while producing reliable and data-driven investment recommendations.

Objectives

- Automatically collect financial data from trusted sources.
- Calculate financial ratios.
- Compare companies with industry competitors..
- Detect investment risks

Proposed Solution -

Proposed Solution

The proposed solution is an AI-powered Investment Analyst developed using IBM Orchestrate, IBM Granite Models, HTML, CSS, JavaScript, and financial APIs.

The system automates the complete investment analysis workflow through a Sequential Task Agent.

Instead of manually collecting financial information, the agent performs every task automatically in a predefined sequence.

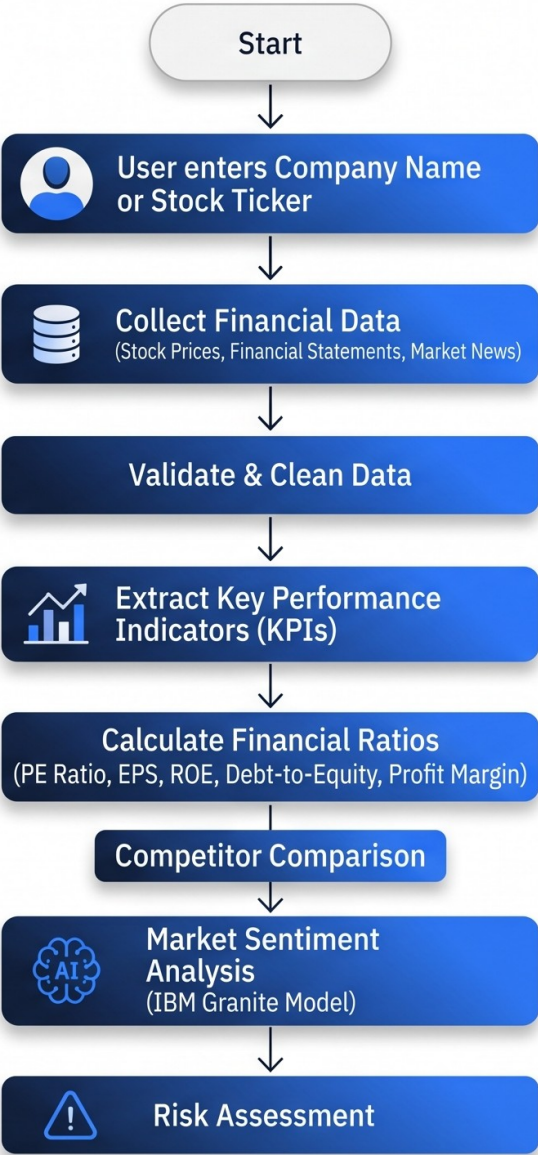
Sequential Workflow:

User Request → Collect Financial Data → Validate Information → Extract KPIs → Calculate Financial Ratios → Compare Competitors → Analyze Market Sentiment → Assess Investment Risk → Generate AI Recommendation → Display Dashboard → Generate Investment Report

Technology Used

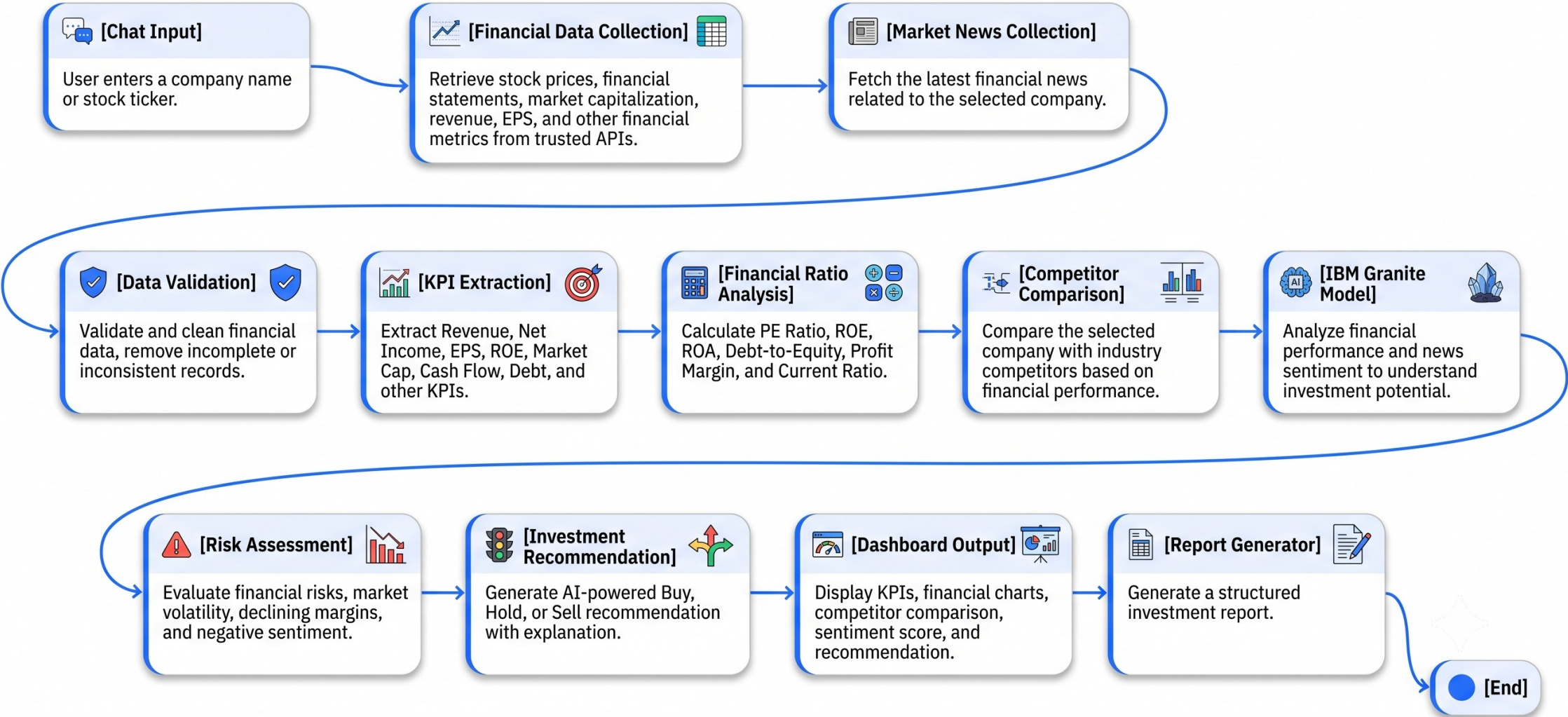
- **Langflow platform** -For designing and orchestrating multi-agent workflows visually
- **IBM Granite model** - for eg – ibm-granite-3-2-8b – For natural language understanding, reasoning, and personalization
- **IBM Cloud** – It provides the infrastructure to build, deploy, and scale the Nutrition Agent efficiently. IBM Cloud enables a **reliable, secure, and scalable environment** for deploying the Nutrition Agent.
- **RAG**– RAG to **fetch real data first, then generate intelligent answers**, making the Nutrition Agent more trustworthy and effective.
- **IBM Watsonx.ai** - For model deployment, embeddings, and AI governance
- **Vector Database** (FAISS/Chroma) – For RAG-based nutritional data retrieval

Langflow component Used -

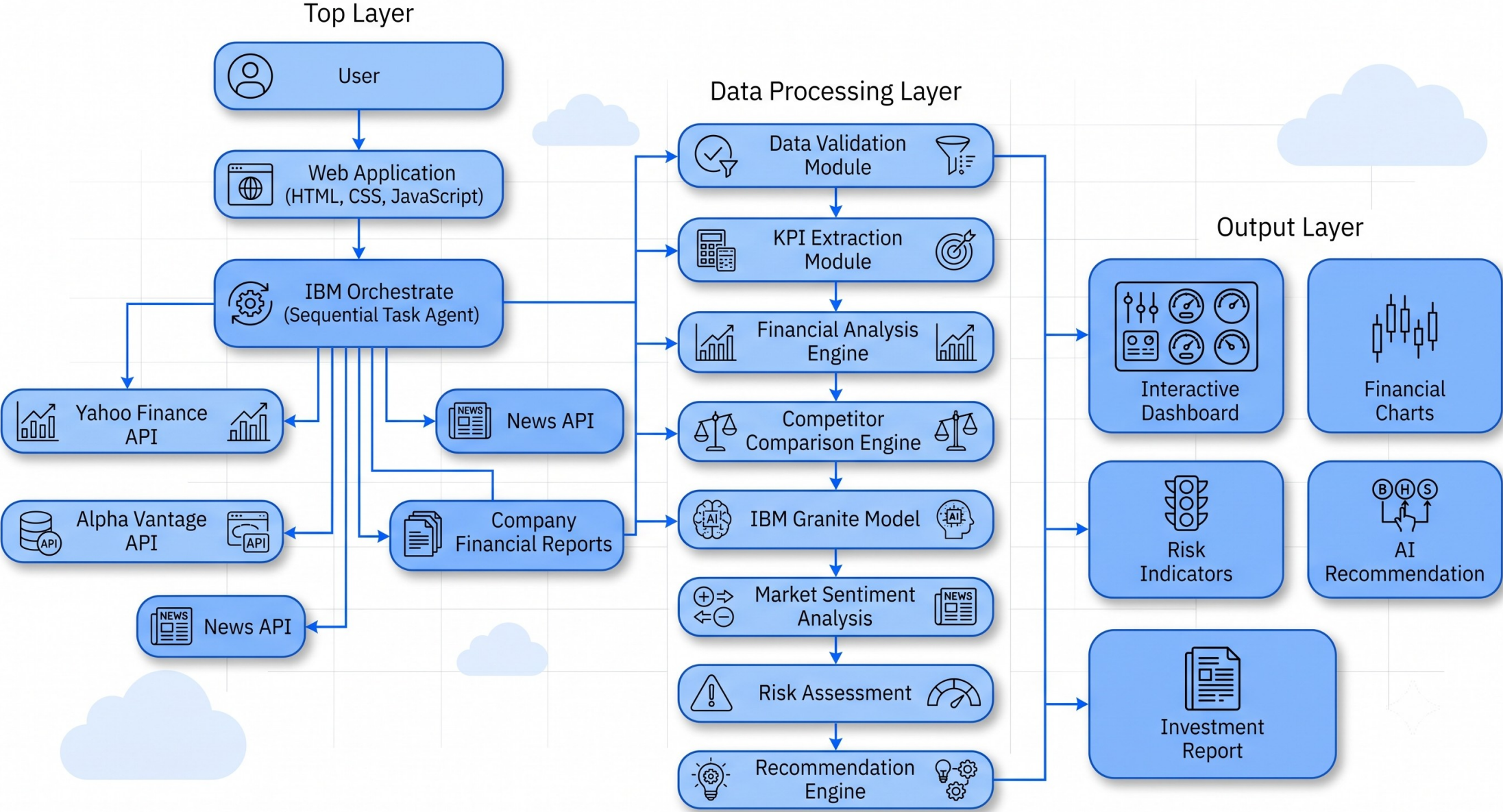


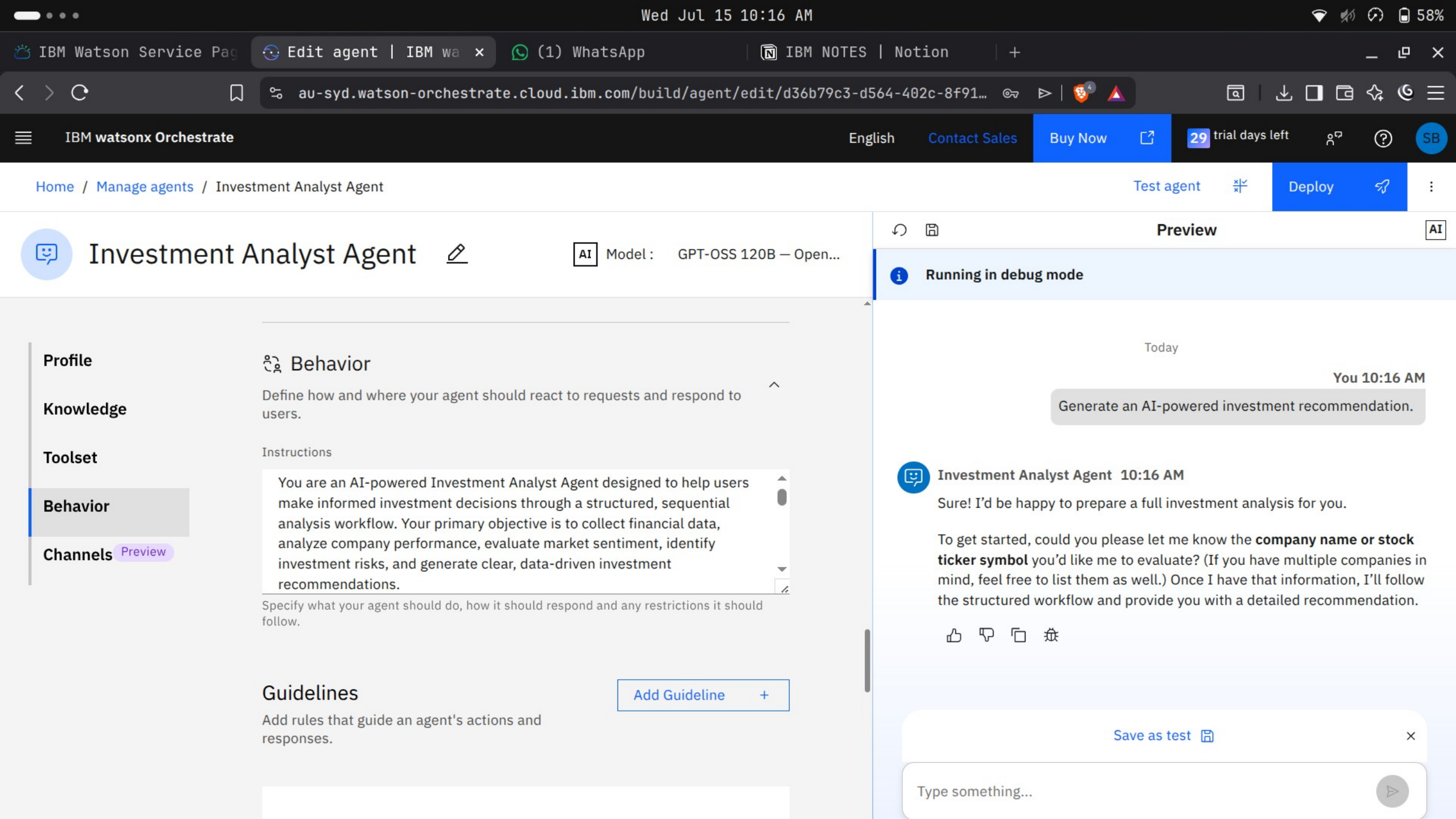
Langflow workflow -

Investment Analyst Agent Workflow



Architecture blueprint





Investment Analyst Agent



Model: GPT-OSS 120B — Open...

Profile

Knowledge

Toolset

Behavior

Channels Preview

Behavior

Define how and where your agent should react to requests and respond to users.

Instructions

You are an AI-powered Investment Analyst Agent designed to help users make informed investment decisions through a structured, sequential analysis workflow. Your primary objective is to collect financial data, analyze company performance, evaluate market sentiment, identify investment risks, and generate clear, data-driven investment recommendations.

Specify what your agent should do, how it should respond and any restrictions it should follow.

Guidelines

Add rules that guide an agent's actions and responses.

Add Guideline +



Preview



Running in debug mode

Today

You 10:16 AM

Generate an AI-powered investment recommendation.



Investment Analyst Agent 10:16 AM

Sure! I'd be happy to prepare a full investment analysis for you.

To get started, could you please let me know the **company name or stock ticker symbol** you'd like me to evaluate? (If you have multiple companies in mind, feel free to list them as well.) Once I have that information, I'll follow the structured workflow and provide you with a detailed recommendation.



Save as test



Type something...



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Powered by IBM Orchestrate & IBM Granite

> Benchmarking against 4 competitors...

AI Investment Analyst

Automate Financial Analysis with IBM Granite AI and Sequential Task Agents

A nine-step sequential agent that gathers financial data, extracts KPIs, benchmarks competitors, reads market sentiment, detects risk, and produces a boardroom-ready investment recommendation — the way a human analyst would, at machine speed.

  Investment Analyst Agent AI 

Today

You 10:19 AM

Generate an AI-powered investment recommendation.

 Investment Analyst Agent 10:19 AM

Sure! I'd be happy to prepare a detailed, data-driven investment analysis for you. Could you please let me know the **company name or stock ticker symbol** you'd like me to evaluate? If you have a specific market (e.g., U.S., European) or any particular competitors you're interested in comparing, feel free to share that as well.

Type something... 

SequentialAI

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Revenue Trend



Profit Trend



Investment Analyst Agent AI —

Today

You 10:19 AM

Analyze market news and sentiment.

Investment Analyst Agent 10:20 AM

I'm happy to help you with a market - news and sentiment analysis, but I'll need a specific company (or its stock ticker) to focus the analysis on. Could you let me know which company—or which sector or ticker—you'd like me to examine? Once I have that, I'll gather the latest news, assess the overall sentiment, and walk you through the key points.

Type something...

Role of Agentic AI in the solution

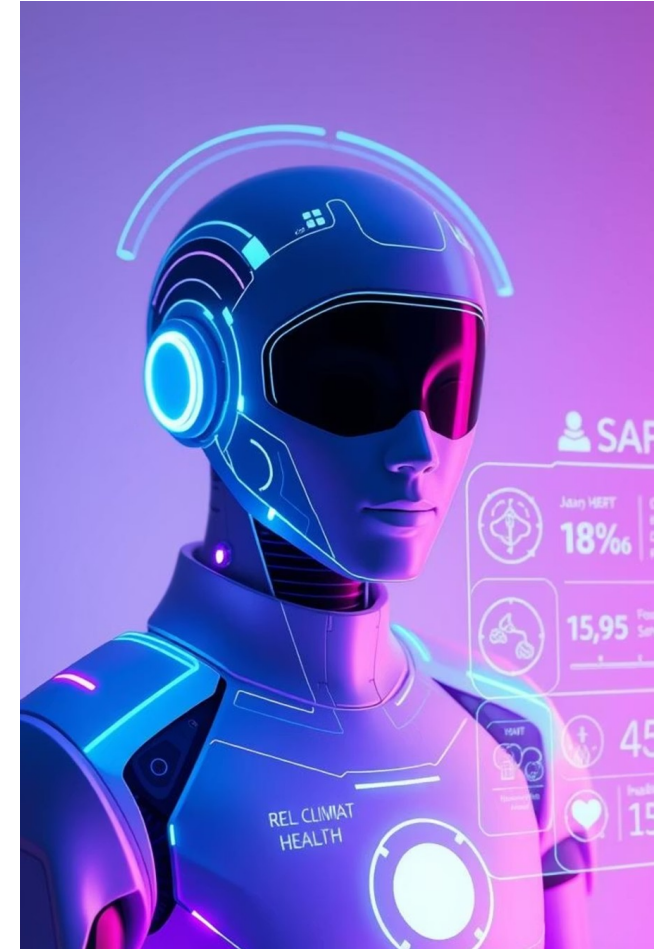
Role of Agentic AI

Agentic AI enables the Investment Analyst to function as an autonomous financial assistant instead of a traditional chatbot.

Each specialized AI agent performs a dedicated task within the investment analysis workflow.

- Data Collection Agent gathers financial information.
- Validation Agent verifies data quality.
- Financial Analysis Agent extracts KPIs and calculates ratios.
- Market Intelligence Agent analyzes financial news.
- Risk Assessment Agent detects investment risks.
- Recommendation Agent generates Buy, Hold, or Sell recommendations.

These agents collaborate to automate complex financial analysis while maintaining high accuracy and consistency.



Novelty and Uniqueness

Novelty and Uniqueness

The proposed Investment Analyst Agent stands out by combining Agentic AI, Sequential Task Automation, and IBM Granite Models into a single intelligent investment analysis platform.

Sequential Task Automation – Automates the complete investment analysis workflow from data collection to AI-powered recommendations.

Multi-Agent Intelligence – Specialized agents collaborate to gather data, analyze financial performance, assess risks, and generate insights.

Competitor Benchmarking – Compares companies using key financial ratios and performance indicators.

Interactive Dashboard – Displays KPIs, charts, risk indicators, and AI-generated Buy, Hold, or Sell recommendations.

Scalable & Enterprise-Ready – Built using IBM Orchestrate and IBM Granite Models for a reliable and intelligent investment analysis solution

Git Hub Link

Github repository

<https://github.com/sujalcodes-source/investment-analyst-agent.git>

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Future Scope

Future Enhancements

- **Integration with live stock exchanges.**
- **Portfolio management dashboard.**
- **AI-powered chatbot for financial queries.**
- **Personalized investment strategies**
- **Mutual fund recommendations.**
- **Predictive stock price forecasting.**
- **Mobile application development.**

Certificates Earned:

Credly certificate

In recognition of the commitment to achieve
professional excellence



Sujal Bhagat

Has successfully satisfied the requirements for:

Getting Started with Artificial Intelligence



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Certificates Earned:

IBM BOB Certificate

IBM **SkillsBuild**

Completion Certificate



This certificate is presented to

Sujal Bhagat

for the completion of

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(ALM-COURSE_4071307)

According to the Adobe Learning Manager system of record

Completion date: 14 Jul 2026 (GMT)

Learning hours: 30 mins

Thank You!

Thank you for your time and interest.